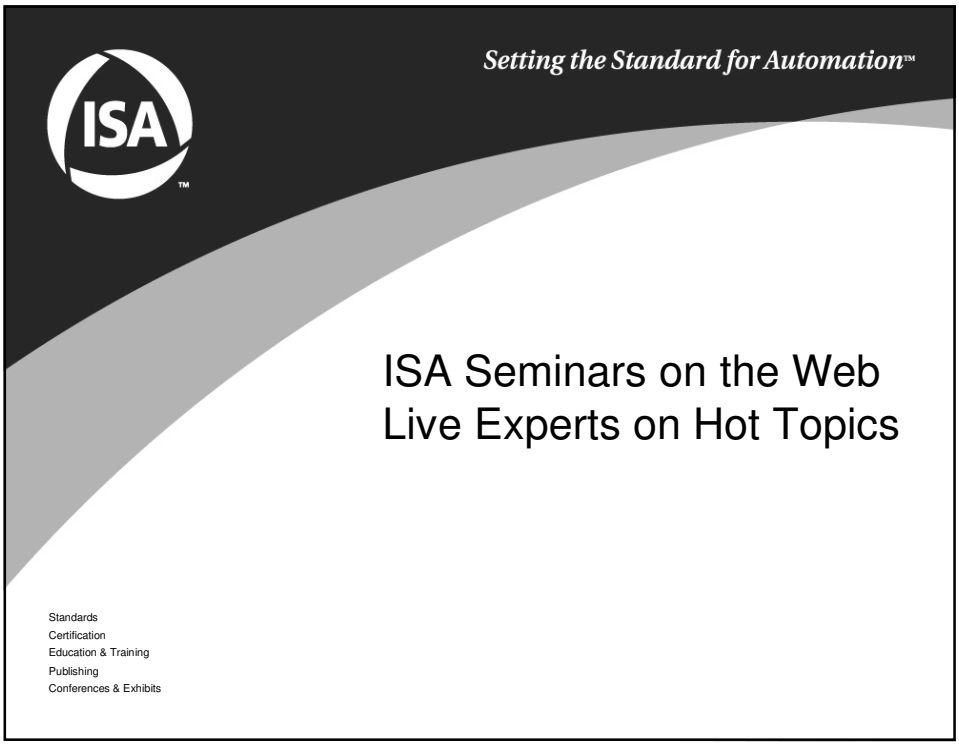
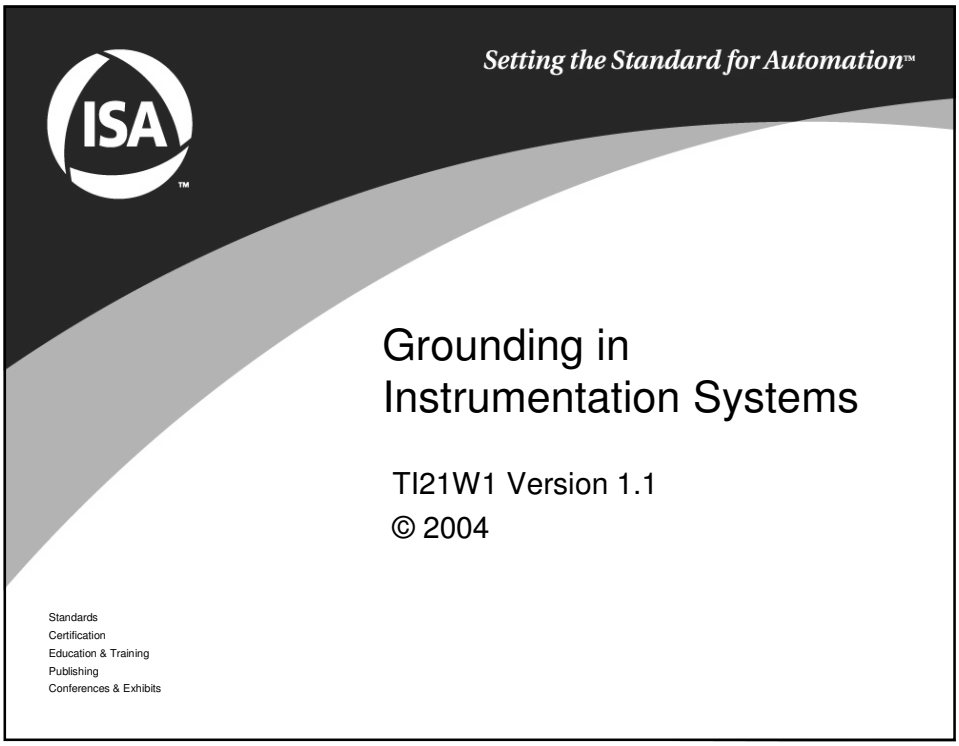

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Setting the Standard for Automation™
Grounding in
Instrumentation Systems
TI21W1 Version 1.1
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Seminar Logistics

- Seminar materials
  - Downloadable presentation
  - Question and Answer session (audio and email)
  - Survey
  - Earn 1 Professional Development Hour (PDH)
- Seminar length
  - 60 minute presentation
  - Three 10-minute question and answer sessions

Audio Instructions

- As a participant, you are in a “listen-only” mode.
- You may ask questions via the internet, using your keyboard, at any time during the presentation. However, the presenter may decide to wait to answer your question until the next Q&A Session.
- If you have audio difficulties, press *0.

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Audio Instructions for Q&A Sessions

- Questions may be asked via your telephone line.
- Press the *1 key on your telephone key-pad.
- If there are no other callers on the line, the operator will announce your name and affiliation to the audience and then ask for your question.
- If other participants are asking questions, you will be placed into a queue until you are first in line.
- While in the queue, you will be in a listen-only mode until the operator indicates that your phone has been activated. The operator will announce your name and affiliation and then ask for your question.

Introduction of ISA Presenter

- ISA Presenter
  - Larry Thompson, CAP
  - ISA Adjunct instructor since 1984. B.A.A.S Tarleton State University. Instructor for Texas State Technical College in Instrumentation (1977-1979), Computer Networking and System Administration (1996-1999), E-Commerce Technology (1999-2003). CCST Certificate, FCC Radiotelephone License (General, formerly First Class). Served 20 years USAF primarily in Electronic Encryption Systems. Owned and managed own services business since 1979.
  - Designed, developed, taught and maintained industrial controls in many varied applications.
  - Author of ISA books: Industrial Data Communications and Basic Electrics/Electronics for Control

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Introduction of Presenter



- Wayne Oswald recently retired after 35 years with The Dow Chemical Company. He served as Maintenance Process Leader for Electrical/Instrumentation/Refrigeration Services functions for Dow's Texas Operations. Since retiring from Dow, Wayne has taught Electrical Continuing Education courses for Associated Builders and Contractors and is currently serving as Project Director for the Greater Houston Construction Careers Initiative, involved in recruiting and training entry level construction workers along the Texas Gulf Coast.
- Wayne received his BS degree in Electrical Engineering Technology from The University of Houston. He serves as Chairman of the Texas Skill Standards Board and is active in The Associated Builders and Contractors and Houston Business Roundtable organizations.

Key Benefits of Seminar

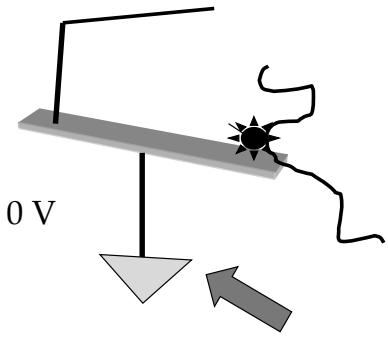
- Know the terminologies surrounding grounds and grounding in Measurement and Control applications.
- Be familiar with the grounding philosophies employed in Measurement and Control applications.
- Understand the concepts of shielding and noise reduction as related to grounding.
- Be able to determine what is the best grounding and shielding approach for different applications.

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Section 1: Electrical Grounds

- What is a ground
- What must be grounded
- What may not be grounded
- Intentional grounding systems
- Separately grounded systems

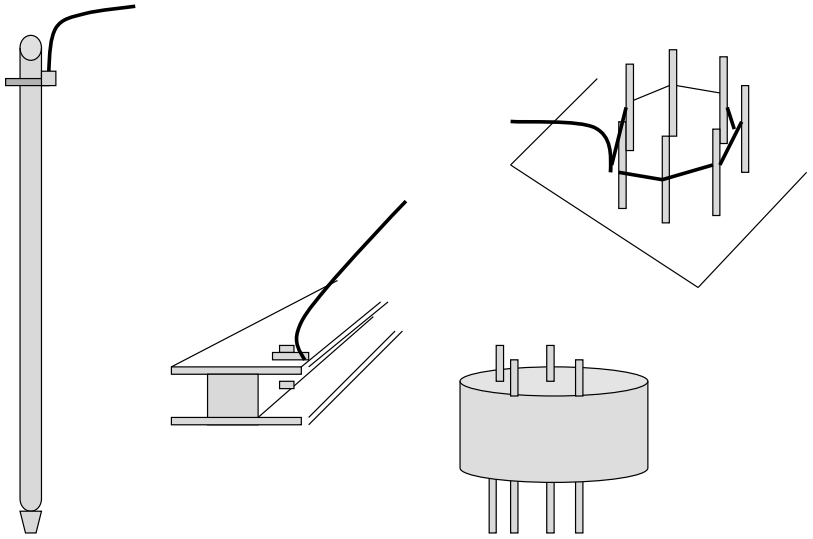
Grounds



Connection to earth or other large conducting mass

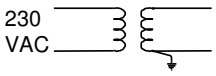
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How do we connect to the “ground” ?

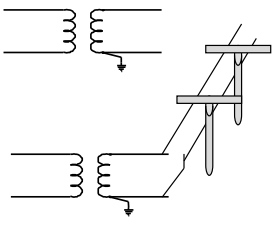


Instrument Power/Signals Requiring Grounding

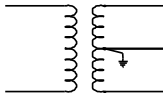
When supplying <50 V

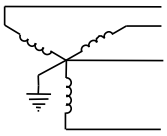


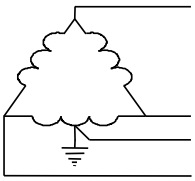
Ungrounded



When supplying 50 – 1KV

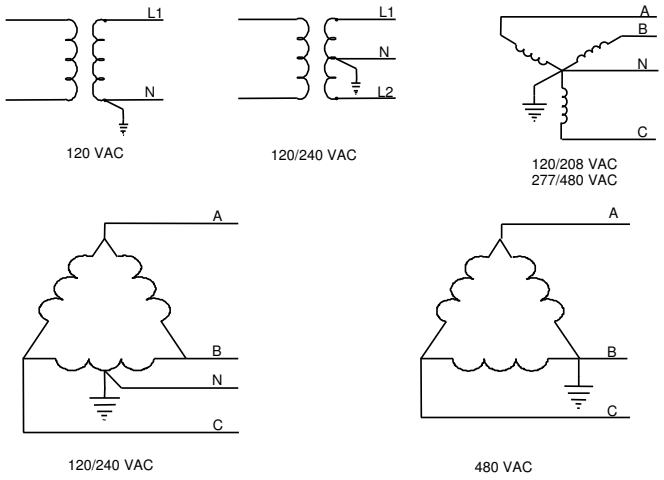






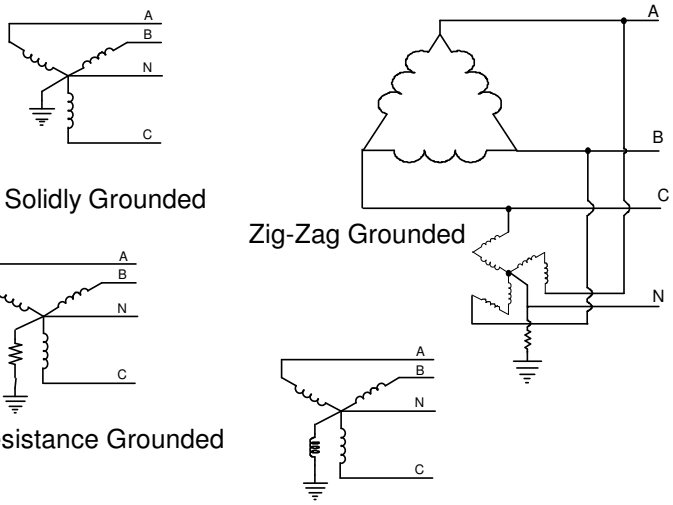
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Intentionally Grounded Systems



Intentionally Grounded Systems

Types of Grounding

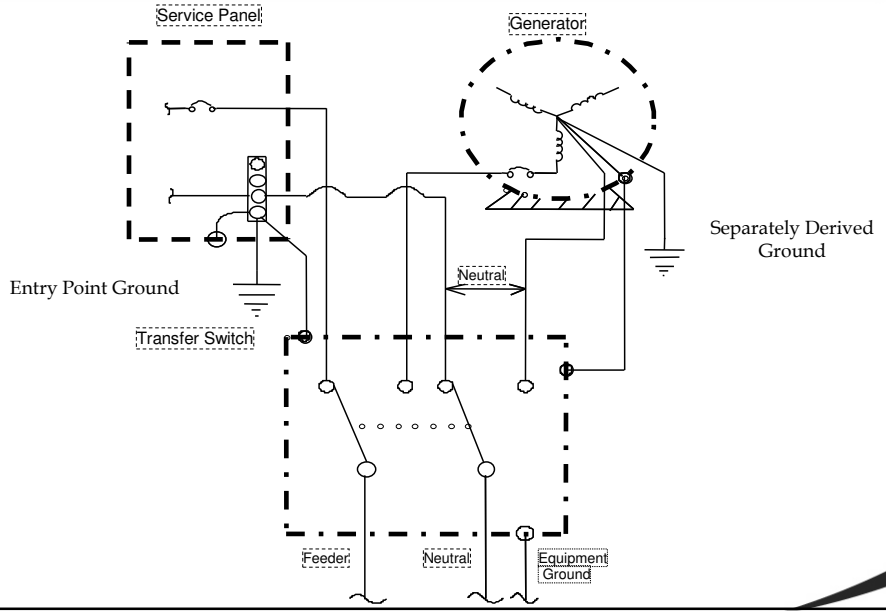


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Instrument Power/Signals Requiring Grounding (cont'd)

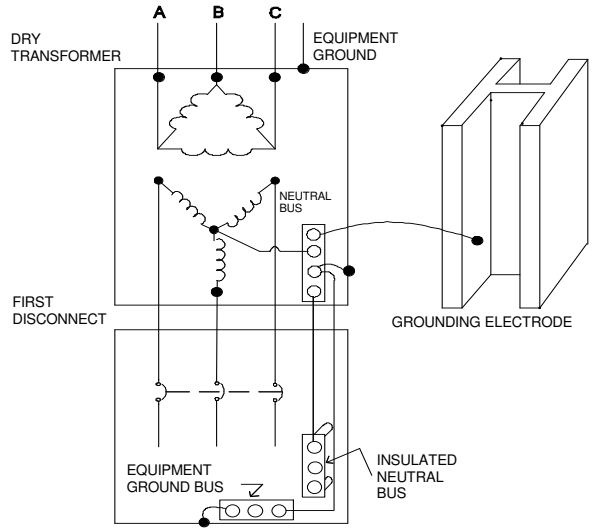
- Separately Derived AC Systems (NEC Article 250-20(d
  - Commonly used in measurement and control circuits
- Direct-Current Circuits: (NEC Article 250-162)
  - Commonly used in measurement and control circuits

Separately Derived System - Generator



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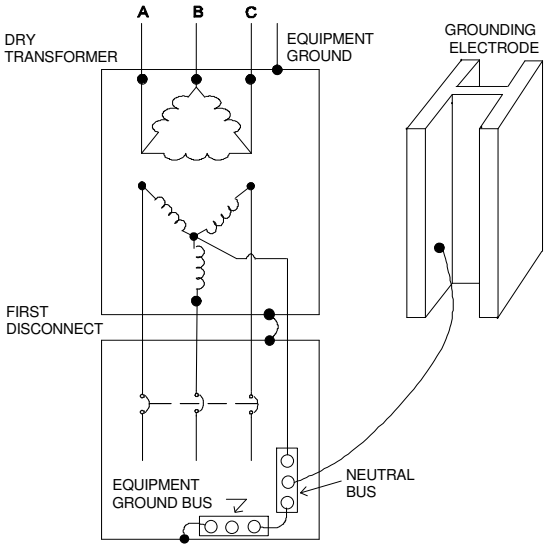
Separately Derived System – Transformer #1



Ground at Transformer

Separately Derived Transformer System
Grounded at the Transformer

Separately Derived Ground – Transformer #2

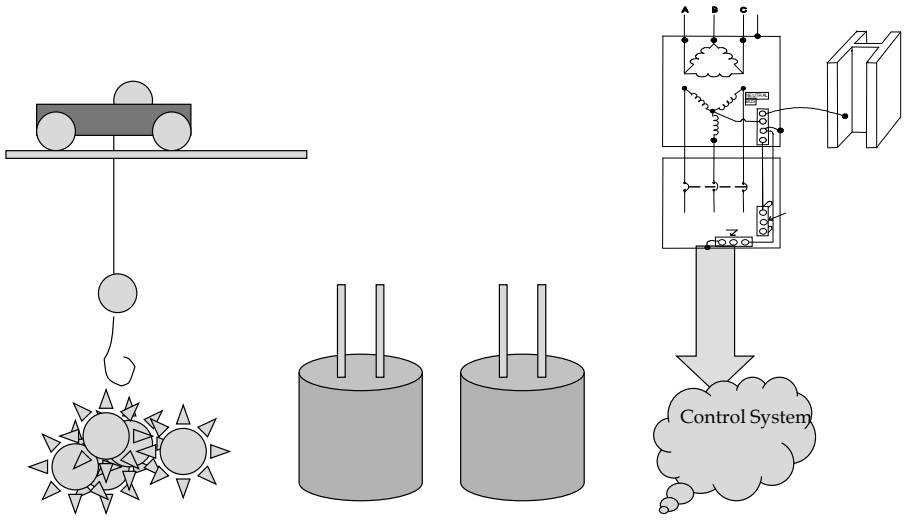


Ground at First Disconnect

Separately Derived Transformer System
Grounded at First Disconnect

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Instrument Power/Signals NOT Requiring Grounding



Review of Key Points

- What is a ground?
- What must be grounded?
- What may not be grounded?
- Intentional grounding systems
- Separately grounded systems

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Live Question and Answer Session

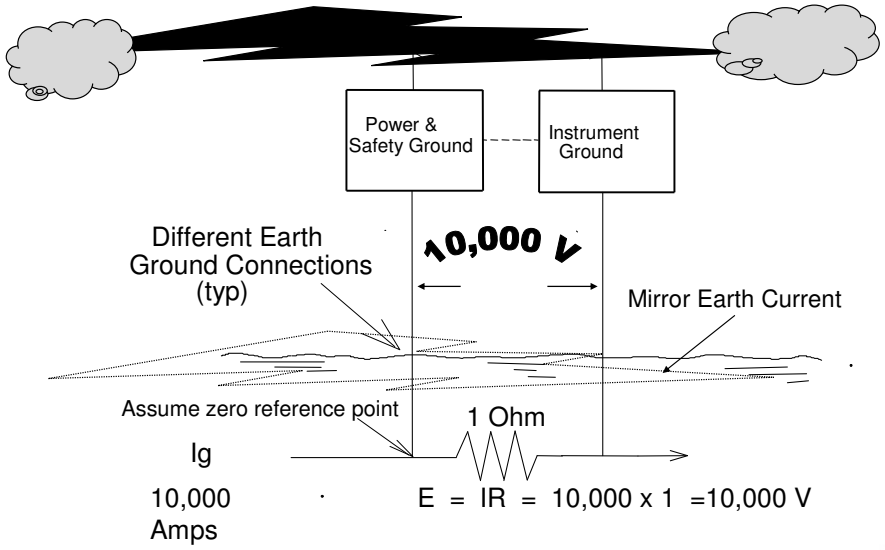
- During Q&A, questions may be asked via your telephone line.
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Section 2: Types of Grounding Systems

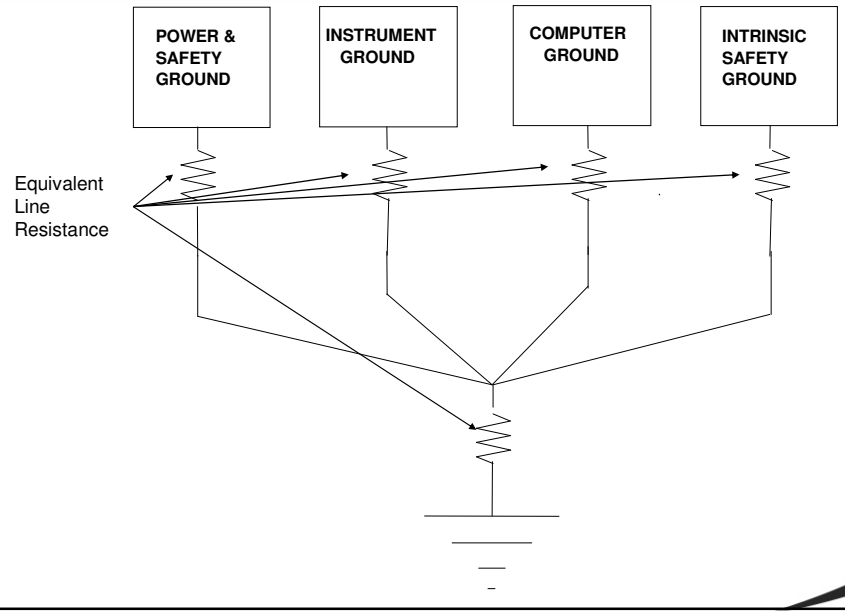
- Ground Considerations
- Single Point Ground (DC to Low Frequency Signals)
- Low Frequency Multi-Point
- High Frequency Single Point Ground
- High Frequency/Low Frequency Model
- Series Ground

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Ground Potential Difference

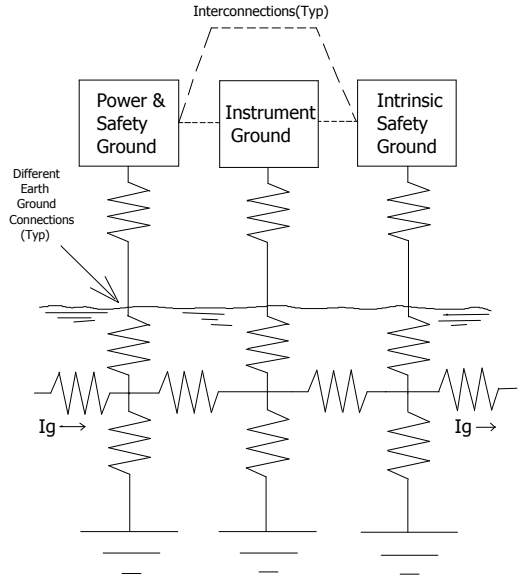


LF Single Point Ground Model

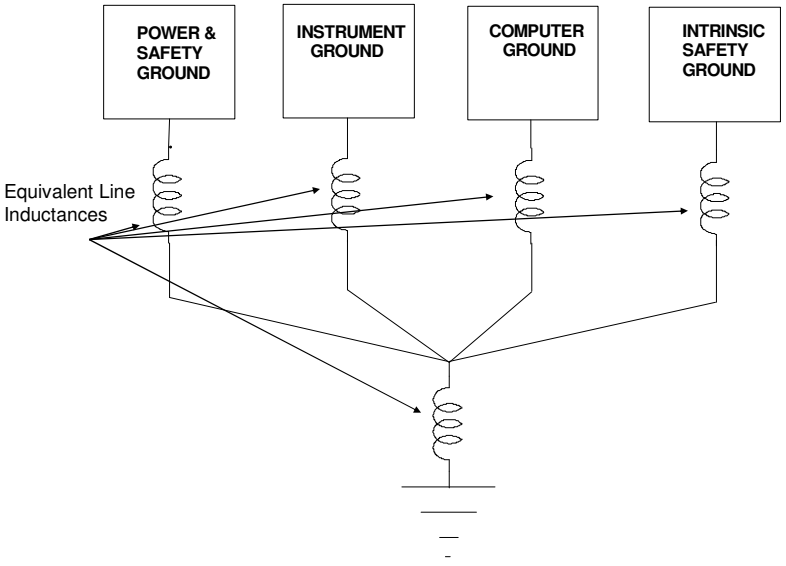


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Low Frequency Multi-Point Ground

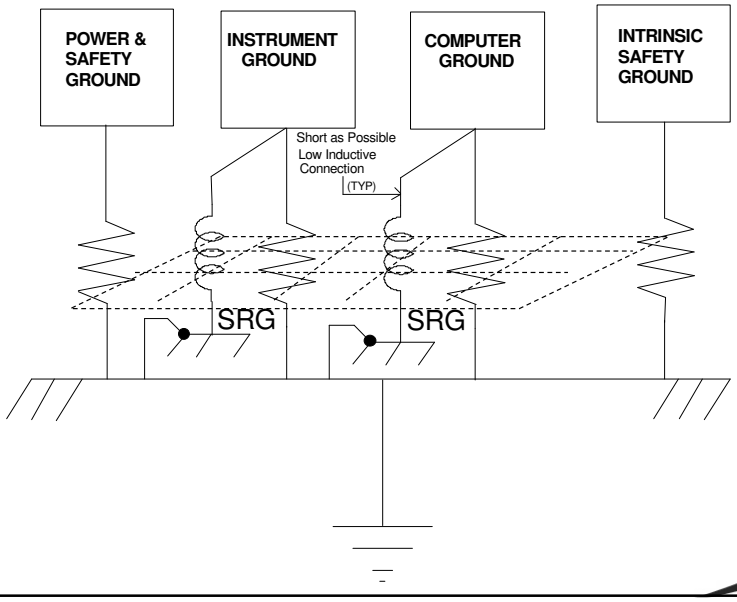


HF Single Single Point Ground Model

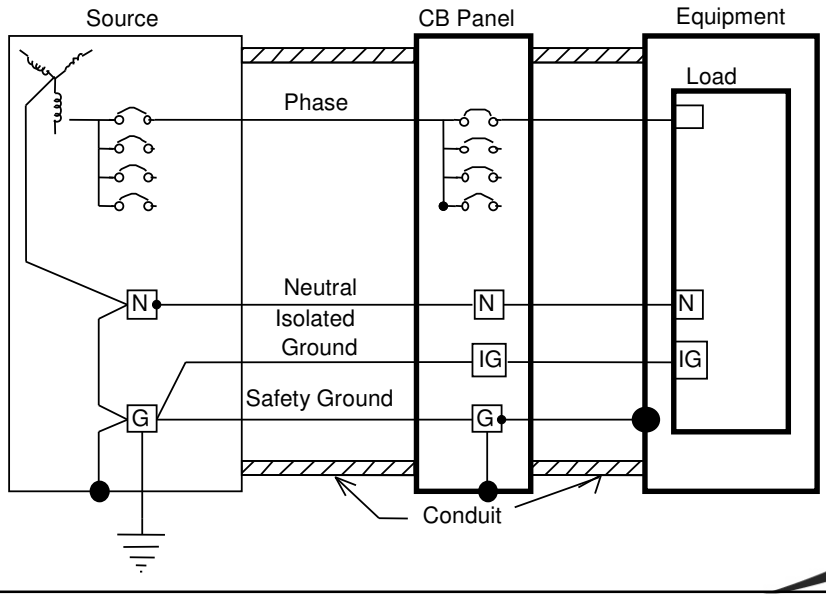


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HF/LF Ground Model

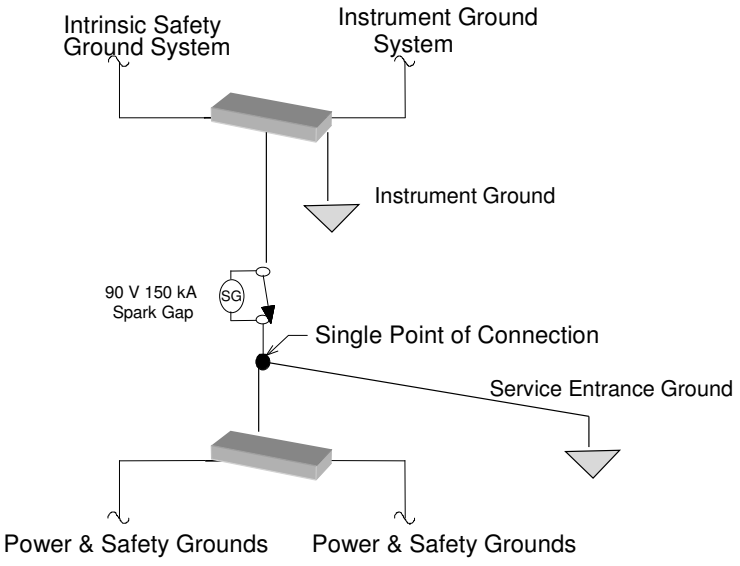


Isolated Ground System(IG)

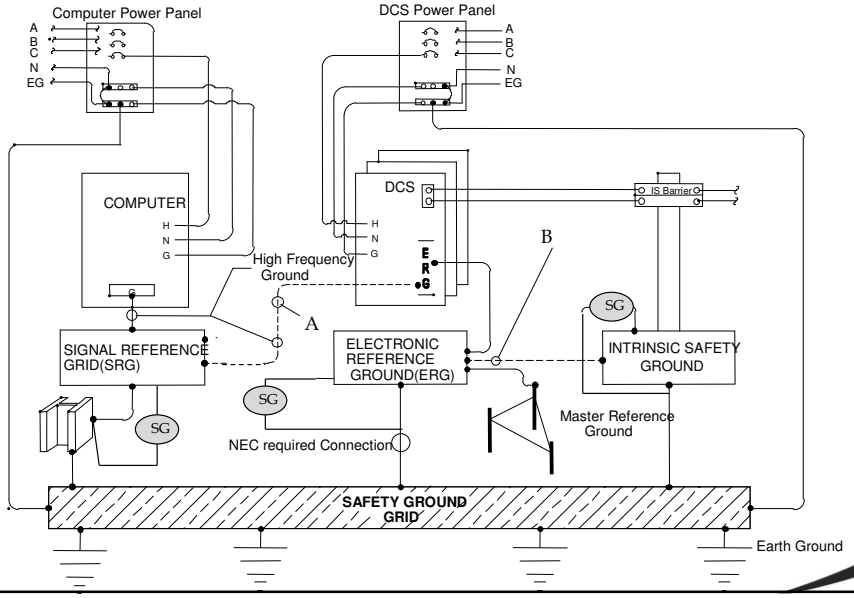


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Simple Illustration of Instrument Grounding System

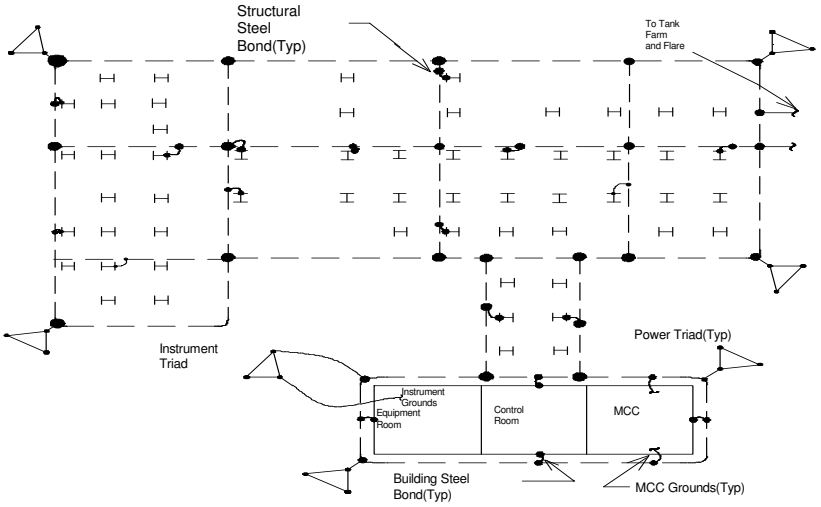


Example DCS Type Grounding



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Example Plant Grid



Example Plant Ground Grid

Review of Key Points

- Ground Considerations
- Single Point Ground (DC to Low Frequency Signals)
- Low Frequency Multi-Point
- High Frequency Single Point Ground
- High Frequency/Low Frequency Model
- Series Ground

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Live Question and Answer Session

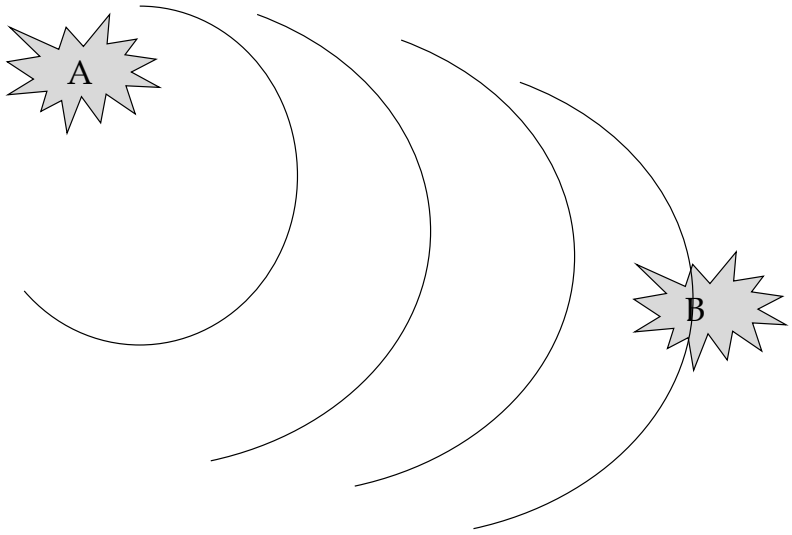
- During Q&A, questions may be asked via your telephone line.
- Press the *1 key on your telephone key-pad.
- If there are no other callers on the line, the operator will announce your name and affiliation to the audience and then ask for your question.
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- While in the queue, you will be in a listen-only mode until the operator indicates that your phone has been activated. The operator will announce your name and affiliation and then ask for your question.

Section 3: Shielding/Noise Reduction

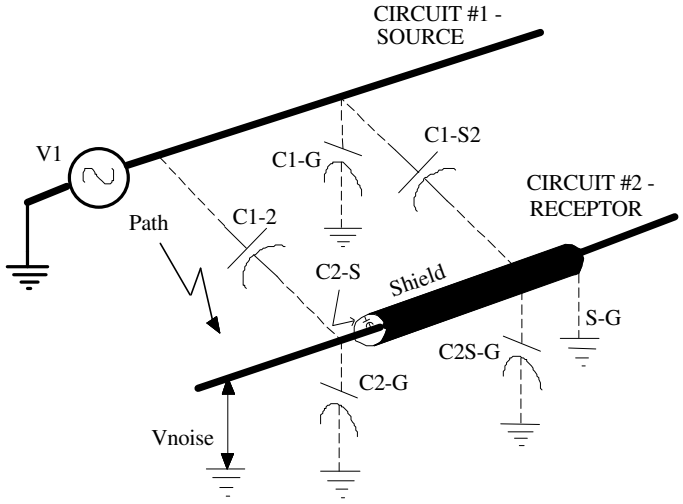
- Field definitions
- Electrostatic Shielding
- Electromagnetic Shielding
- Shielding Applications

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Field Definitions



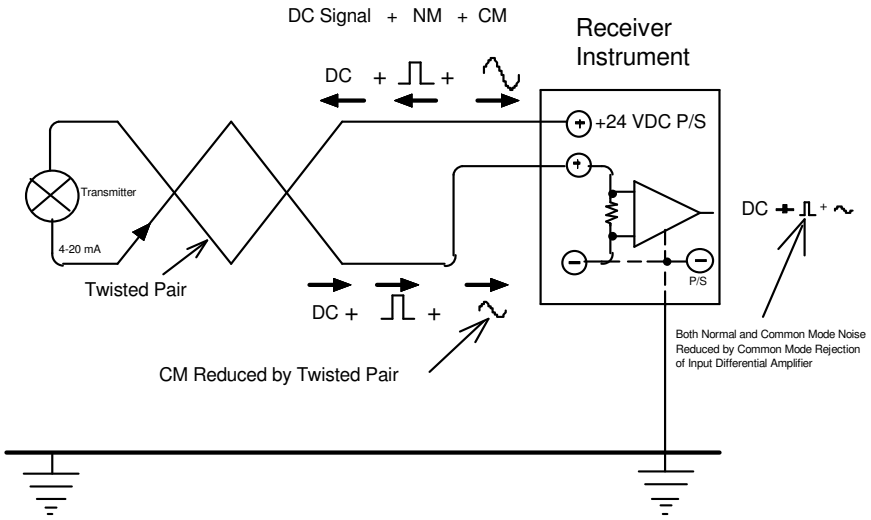
Electrostatic Shield Model



Electrostatic Shield Model (Capacitive)

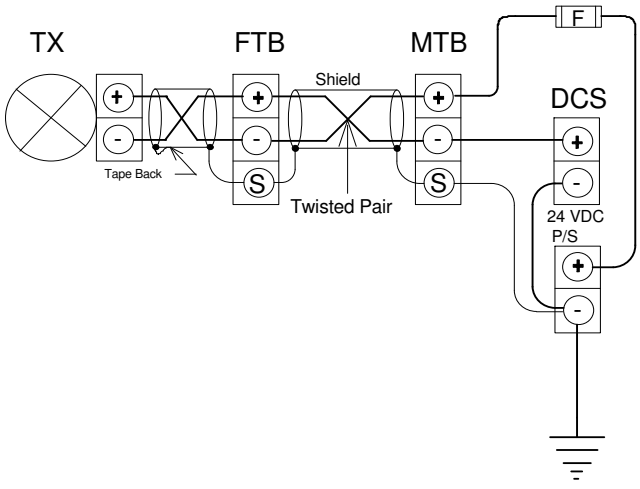
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Noise in a 4-20 mA Loop



Noise in a 4-20 mA Loop

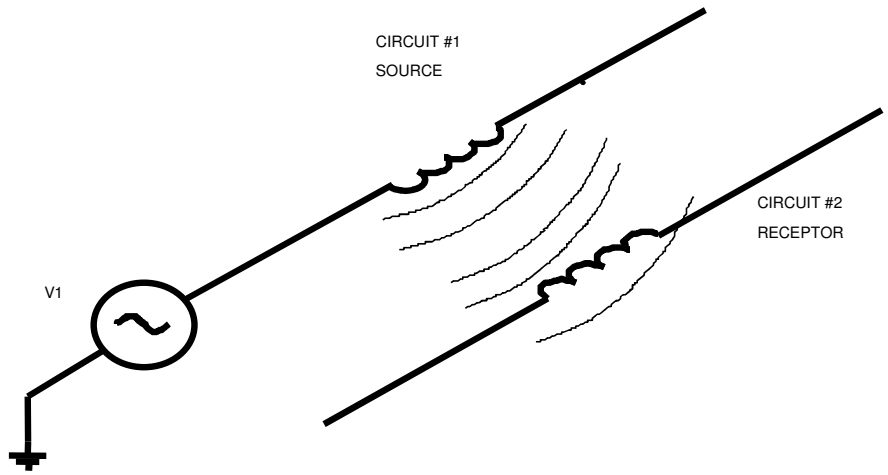
4-20 mA Instrument Cable Shield



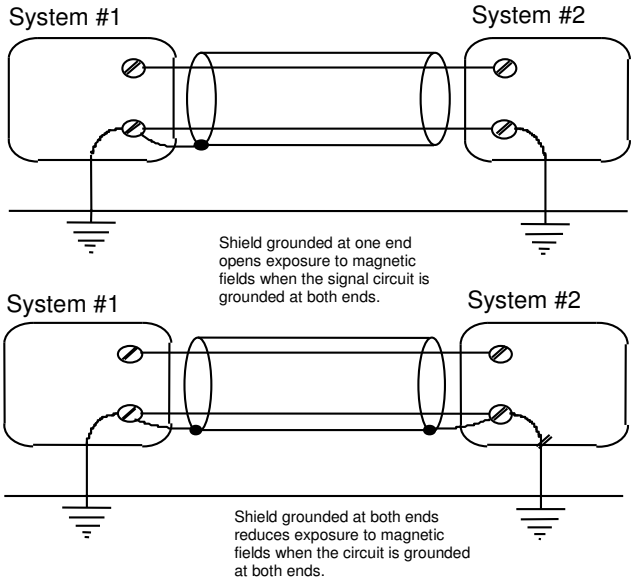
4-20 mA Instrument Loop Shield

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Magnetic Coupling Model

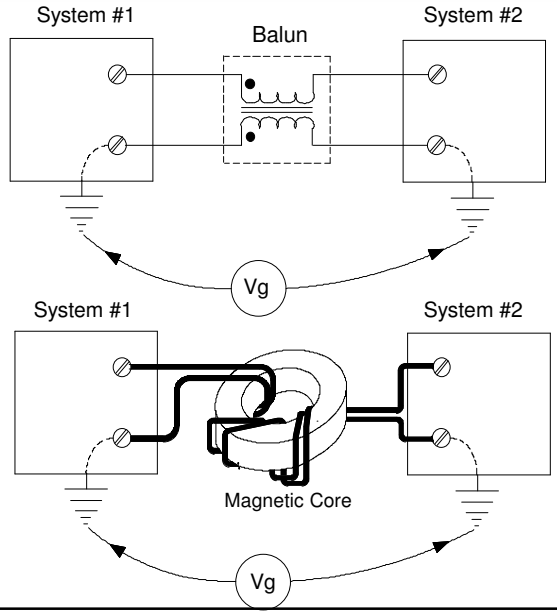


Signal Circuits Grounded at Both Ends

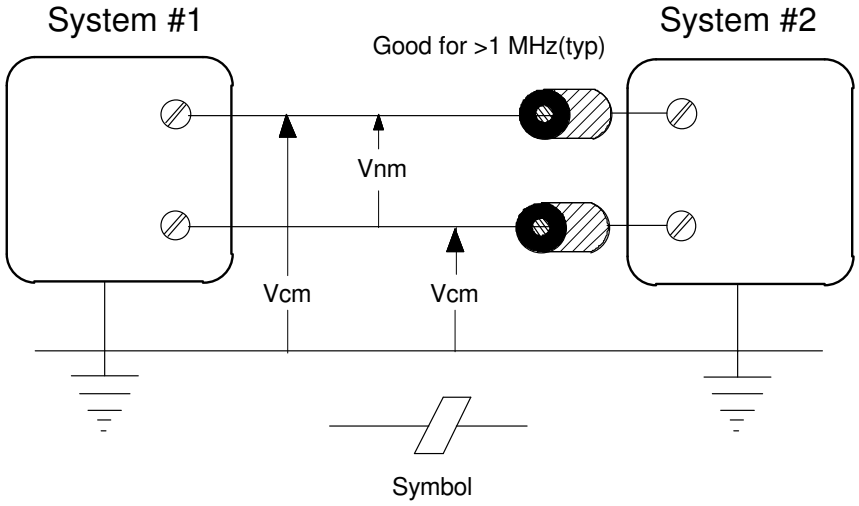


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Common Mode Chokes

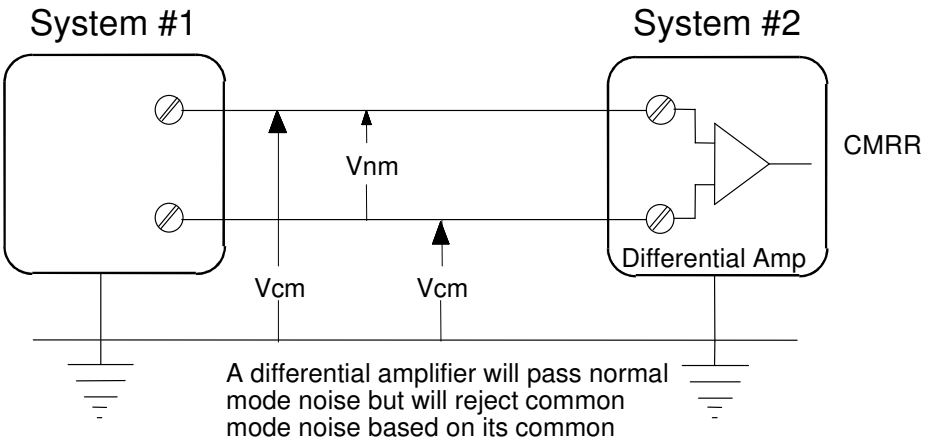


Ferrite Beads



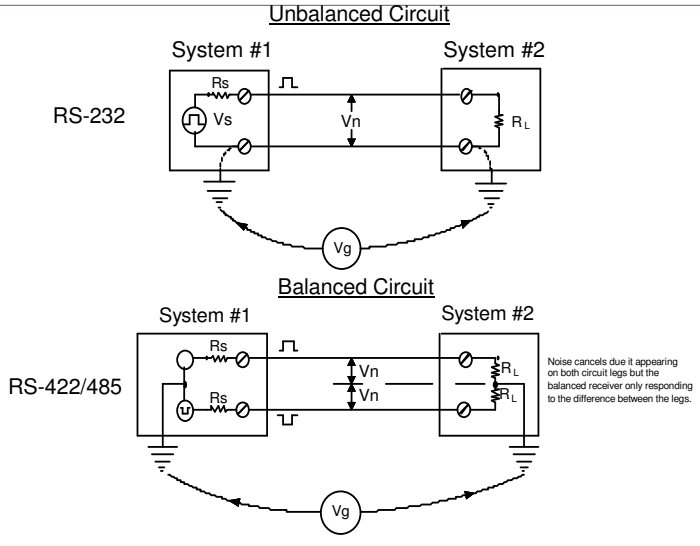
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Differential Amplifier



A differential amplifier will pass normal mode noise but will reject common mode noise based on its common mode reduction ratio (CMRR).

Balanced Signal Lines



Balanced and Unbalanced Lines

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Review of Key Points

- Field definitions
- Electrostatic Shielding
- Electromagnetic Shielding
- Shielding Applications

How Many People are at Your Site?

- Poll Slide
- Click on the appropriate number indicating the number of people that are at your site.

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Live Question and Answer Session

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Related Courses from ISA

- Grounding and Noise Considerations for Control Equipment and Computers (TI21)
- Understanding Electrical Systems (TI15)
- Industrial Electronics (TI20)
- Electrical Noise Reduction in Instrumentation Systems (TI21W2)

- All ISA courses are available any time as on-site training
- For more information: **www.isa.org/training** or **(919) 549-8411**

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Other Related Resources from ISA

- *Electrical Instruments in Hazardous Locations, 4th Edition* by E.C. Magison from ISA Press
- The ANSI/ISA12 series of standards and technical reports on electrical instruments & apparatus
- ISA Membership connects you to people and ideas in automation and control
  - Just $85 per year, plus $9 for each division
- For more information: **www.isa.org/shopISA** or **(919) 549-8411**

ISA Certifications

- Certified Automation Professionals ® (CAP ®)
  - www.isa.org/CAP
- Certified Control Systems Technician® (CCST®)
  - www.isa.org/CCST
- Please visit us online for more information on any of these programs, or call (919) 549-8411.

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Seminar Survey

- Web-based Seminar Surveys are tools to help ISA maintain the quality of our training programs.
- Please complete the following four slides of the survey and fax to ISA when it is complete.

Please fax completed survey pages to (919) 549-8288.

Seminar Survey (cont'd)

Date of Web-Seminar _____

Please mark only one circle for each question.

Scale Definition: P-Poor; F-Fair; G-Good; VG-Very Good; E-Excellent

	P	F	G	VG	E
1. Seminar objectives were clear and attainable with the allotted time.	○	○	○	○	○
2. Pre-program seminar description was accurate.	○	○	○	○	○
3. How was registration process?	○	○	○	○	○
4. How was the audio quality?	○	○	○	○	○

Please fax completed survey pages to (919) 549-8288.

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Seminar Survey (cont'd)

Name of Presenter: _____

	P	F	G	VG	E
5. Demonstrated practical knowledge about the subject.	◯	◯	◯	◯	◯
6. Provided clear answers to questions/comments.	◯	◯	◯	◯	◯

7. How many people listened at your site?

◯ 1-5 ◯ 6-10 ◯ 11-15 ◯ 16-20 ◯ 21+

Please fax completed survey pages to (919) 549-8288.

Seminar Survey (cont'd)

8. Would you participate in another virtual seminar? Yes ◯ No ◯

9. Did the web component add to the value of the virtual seminar? Yes ◯ No ◯

10. Did the seminar satisfy your professional objectives? Yes ◯ No ◯

11. How did you hear about this seminar?

_____

_____

Please fax completed survey pages to (919) 549-8288.

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Seminar Survey (cont'd)

12. List other topics you would like to view as a web seminar, along with your overall impression of the program.

_____

_____

13. May we quote your comments?

○ **Yes** ○ **No**

Name of Participant (optional) _____

Name of Your Company _____

Please fax completed survey pages to (919) 549-8288.

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